

2. What is the term a_8 of the sequence $\{a_n\}$ if a_n equals
- a) 2^{n-1} ? b) 7^n ?
c) $1 + (-1)^n$? d) $-(-2)^n$?
3. What are the terms a_0, a_1, a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals
- a) $2^n + 1$? b) $(n + 1)^{n+1}$?
c) $\lfloor n/2 \rfloor$? d) $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?

$$a) \quad a_8 = 2^{8-1} = 2^7$$

$$b) \quad a_8 = 7$$

$$c) \quad a_8 = 1$$

$$d) \quad a_8 = -2^8$$

3)

$$a) \quad a_0 = 2, a_1 = 3, a_2 = 5, a_3 = 7$$

$$b) \quad a_0 = 1, a_1 = 9, a_2 = 27, a_3 = 4^9$$

$$c) \quad a_0 = 0, a_1 = 1/2, a_2 = 1, a_3 = 3/2$$

$$d) \quad a_0 = 0, a_1 = 1, a_2 = 2, a_3 = 3$$

10. Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.

- a) $a_n = -2a_{n-1}, a_0 = -1$
b) $a_n = a_{n-1} - a_{n-2}, a_0 = 2, a_1 = -1$
c) $a_n = 3a_{n-1}^2, a_0 = 1$
d) $a_n = na_{n-1} + a_{n-2}^2, a_0 = -1, a_1 = 0$
e) $a_n = a_{n-1} - a_{n-2} + a_{n-3}, a_0 = 1, a_1 = 1, a_2 = 2$

$$a) \quad a_0 = -1, a_1 = 2, a_2 = -4, a_3 = 8, a_4 = -16, a_5 = 32$$

$$b) \quad a_0 = 2, a_1 = -1, a_2 = -3, a_3 = -2, a_4 = 1, a_5 = 3$$

$$c) \quad a_0 = 1, a_1 = 3, a_2 = 3^3, a_3 = 3^7, a_4 = 3^{15}, a_5 = 3^{31}$$

$$d) \quad a_0 = -1, a_1 = 0, a_2 = 1, a_3 = 3, a_4 = 13, a_5 = (5)(13) + 9$$

$$e) \quad a_0 = 1, a_1 = 2, a_2 = 2, a_3 = 1, a_4 = 1, a_5 = 1$$

13. Is the sequence $\{a_n\}$ a solution of the recurrence relation

$$a_n = 8a_{n-1} - 16a_{n-2} \text{ if}$$

- a) $a_n = 0$? b) $a_n = 1$?
c) $a_n = 2^n$? d) $a_n = 4^n$?
e) $a_n = n4^n$? f) $a_n = 2 \cdot 4^n + 3n4^n$?
g) $a_n = (-4)^n$? h) $a_n = n^24^n$?

$$a) \quad \text{yes}$$

$$b) \quad \text{no}$$

$$c) \quad a_0 = 1, a_1 = 2, a_2 = 4 \quad f(2) - 16(1) = 0 \quad \text{Not solution}$$

$$d) \quad a_0 = 1, a_1 = 4, a_2 = 16 \quad f(4) - 16(2) = 16 \quad \text{yes solution}$$

$$e) \quad a_0 = 0, a_1 = 4, a_2 = 32 \quad f(4) - 16(0) = 32 \quad \text{yes solution}$$

$$f) \quad a_0 = 2, a_1 = 20, a_2 = 128 \quad f(20) - (16)(2) = 160 - 32 = 128 \quad \text{yes solution}$$

$$g) \quad a_0 = 1, a_1 = -4, a_2 = 16 \quad f(-4) - 16 = -48 \quad \text{Not a solution}$$

$$h) \quad a_0 = 0, a_1 = 4, a_2 = 64 \quad f(4) - 16(0) = 32 \quad \text{Not a solution}$$

16. Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach such as that used in Example 10.

- a) $a_n = -a_{n-1}, a_0 = 5$
b) $a_n = a_{n-1} + 3, a_0 = 1$
c) $a_n = a_{n-1} - n, a_0 = 4$
d) $a_n = 2a_{n-1} - 3, a_0 = -1$
e) $a_n = (n + 1)a_{n-1}, a_0 = 2$
f) $a_n = 2na_{n-1}, a_0 = 3$
g) $a_n = -a_{n-1} + n - 1, a_0 = 7$

$$a) \quad a_0 = 5, a_1 = -5, a_2 = 5 \quad a_n = 5(-1)^n$$

$$b) \quad a_0 = 1, a_1 = 4, a_2 = 7 \quad a_n = 1 + 3n$$

$$c) \quad a_0 = 4, a_1 = 3, a_2 = 1, a_3 = -2 \quad a_n = 4 - \frac{n(n+1)}{2}$$

$$d) \quad a_0 = 1, a_1 = 5, a_2 = -13, a_3 = -29 \quad a_n = -2^{n+2} + 3$$

$$e) \quad a_0 = 2, a_1 = 4, a_2 = 12, a_3 = 48 \quad a_n = 2(n+1)!$$

$$f) \quad a_0 = 3, a_1 = 6, a_2 = 24, a_3 = 144 \quad a_n = 32^n n!$$

$$g) \quad a_0 = 7, a_1 = -7, a_2 = 8, a_3 = -6 \quad a_n =$$

What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

- a) $\sum_{j \in S} j$
- b) $\sum_{j \in S} j^2$
- c) $\sum_{j \in S} (1/j)$
- d) $\sum_{j \in S} 1$

- a) 16
- b) 84
- c) $1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{7}$
- d) 4

33. Compute each of these double sums.

- a) $\sum_{i=1}^2 \sum_{j=1}^3 (i+j)$
- b) $\sum_{i=0}^2 \sum_{j=0}^3 (2i+3j)$
- c) $\sum_{i=1}^3 \sum_{j=0}^2 i$
- d) $\sum_{i=0}^2 \sum_{j=1}^3 ij$

- a) $9 + 12 = 21$
- b) $18 + 24 + 36 = 78$
- c) $3 + 6 + 9 = 18$
- d) $6 + 12 = 18$

$$\{5\}$$

4. Find $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if f is defined recursively by $f(0) = f(1) = 1$ and for $n = 1, 2, \dots$

- a) $f(n+1) = f(n) - f(n-1)$.
- b) $f(n+1) = f(n)f(n-1)$.
- c) $f(n+1) = f(n)^2 + f(n-1)^3$.
- d) $f(n+1) = f(n)/f(n-1)$.

- a) $f(2) = 0, f(3) = -1, f(4) = -1, f(5) = 0$
- b) $f(2) = 1, f(3) = 1, f(4) = 1, f(5) = 1$
- c) $f(2) = 2, f(3) = 5, f(4) = 33, f(5) = 33^2 + 25^3$
- d) $f(2) = 1, f(3) = 1, f(4) = 1, f(5) = 1$

5. Determine whether each of these proposed definitions is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers. If f is well defined, find a formula for $f(n)$ when n is a nonnegative integer and prove that your formula is valid.

- a) $f(0) = 0, f(n) = 2f(n-2)$ for $n \geq 1$
- b) $f(0) = 1, f(n) = f(n-1) - 1$ for $n \geq 1$
- c) $f(0) = 2, f(1) = 3, f(n) = f(n-1) - 1$ for $n \geq 2$
- d) $f(0) = 1, f(1) = 2, f(n) = 2f(n-2)$ for $n \geq 2$
- e) $f(0) = 1, f(n) = 3f(n-1)$ if n is odd and $n \geq 1$ and $f(n) = 9f(n-2)$ if n is even and $n \geq 2$

- a) Not valid
- b) $f(n) = 1 - n$ for $n \geq 1$
- c) $f(n) = 4 - n$ for $n \geq 2$
- d) $f(n) = 2^{\lceil n/2 \rceil}$ for $n \geq 2$
- e) $f(n) = 3^n$ for $n \geq 1$

8. Give a recursive definition of the sequence $\{a_n\}$, $n = 1, 2, 3, \dots$ if

- a) $a_n = 4n - 2$.
- b) $a_n = 1 + (-1)^n$.
- c) $a_n = n(n+1)$.
- d) $a_n = n^2$.

- a) $f(1) = 2, f(2) = 6, f(3) = 10, f(n) = f(n-1) + 4, f(1) = 0$ for $n \geq 1$
- b) $a_1 = 0, f(n) = a_{n-1} + 2(-1)^n$ for $n \geq 1$
- c) $a_1 = 2, f(n) = a_{n-1} + 2n, f(1) = 2, f(2) = 6, f(3) = 22, f(4) = 20$
- d) $a_1 = 1, f(n) = a_{n-1} + 1 + 2(n-1)$ for $n \geq 1, f(1) = 1, f(2) = 4, f(3) = 9, f(4) = 16$

22. Show that the set S defined by $1 \in S$ and $s + t \in S$ whenever $s \in S$ and $t \in S$ is the set of positive integers.

to prove that the recursive definition you found is correct. [Hint: To find a recursive definition, plot the points in the set in the plane and look for patterns.]

- a) $S = \{(a, b) \mid a \in \mathbb{Z}^+, b \in \mathbb{Z}^+, \text{ and } a + b \text{ is even}\}$
- b) $S = \{(a, b) \mid a \in \mathbb{Z}^+, b \in \mathbb{Z}^+, \text{ and } a \text{ or } b \text{ is odd}\}$
- c) $S = \{(a, b) \mid a \in \mathbb{Z}^+, b \in \mathbb{Z}^+, a + b \text{ is odd, and } 3 \mid b\}$

$$s, t = 1 \quad 1+1=2$$

$$s=2 \quad t=1 \quad 1+2=3$$

$$a). \text{ in finitum.}$$

23. Give a recursive definition of the set of positive integers that are multiples of 5.

24. Give a recursive definition of

- a) the set of odd positive integers.
- b) the set of positive integer powers of 3.
- c) the set of polynomials with integer coefficients.

23) $a_1 = 0 \quad \{a \mid a_n = a_{n-1} + 5\} \text{ for } n > 1$

24)
a) $a_1 = 1 \quad \{a \mid a_n = a_{n-1} + 2\} \text{ for } n > 1$

b) $a_1 = 3 \quad \{a \mid a_n = (a_{n-1})^2\} \text{ for } n > 1$

c)